Sahaj Bhadja

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Summary

Software Engineer with 2+ years shipping **backend services**, **REST APIs**, and **Azure**-hosted microservices, building **C#/.NET Core** and **SQL Server**-backed systems with a React frontend. Strong OOP foundation across Java, Python, and TypeScript.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected multi-tenant backend services on **Azure SQL** and **SQL Server** with row-level security and normalized schemas for 500+ users, tuning indexed queries that cut dashboard latency from 1.2s to 350ms.
- Built a fault-tolerant async orchestration service with FastAPI background workers and **CI/CD**-backed releases, processing 1K+ jobs with explicit state transitions and failure isolation, tripling throughput under concurrent load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Designed **React** components and **REST API** endpoints for an AI grading platform, integrating Flask backend services on AWS RDS/MySQL and shipping features via GitHub Actions **CI/CD** pipelines for 150+ instructors.
- Built a concurrent batch-processing pipeline with multithreading and **REST APIs**, handling 10K+ daily requests while maintaining stable API performance and reducing feedback turnaround time by 60% under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Maintained 20+ automated test suites with Jenkins, PyTest, and Mockito across a **microservice** architecture, validating REST API integrations and cutting regressions while boosting **CI** pipeline efficiency by 68%.
- Optimized large-scale API integrations using AWS Glue and S3 in an **Agile** Scrum environment, improving response times 30% and cutting technical debt 20% across release pipelines efficiently.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Deployed a MERN-based **microservices** platform to **Azure App Services** with **Azure SQL** persistence and JWT authentication, automating Jenkins **CI/CD** and Docker-based packaging across staged production releases.
- Built real-time WebSocket-based CRM features syncing 2M+ records across 8 microservices, monitoring performance with Azure Monitor and Sentry while reducing resource wastage 42% and improving lead conversion 15%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Designed the end-to-end architecture for a cross-platform booking application, building Flask **REST APIs** and Firebase-backed real-time sync that delivered sub-200ms latency across 50K+ booking records on Android, iOS, and web.
- Implemented coupon validation, payment webhook handling, and wallet-signature **authentication**, reducing manual refund handling by roughly 70% while maintaining PCI-compliant multi-currency payment workflows.

Skills

Languages: C#, Python, Java, JavaScript/TypeScript, SQL, Bash/Shell
Backend & APIs: .NET Core/.NET 5+, ASP.NET Core Web API, Entity Framework Core, REST APIs, Flask, FastAPI, Node.js
Frontend: React.js, TypeScript, HTML, CSS
Databases: SQL Server, PostgreSQL, MySQL, Azure SQL, MongoDB
Cloud & Infrastructure: Microsoft Azure (App Services, AKS, SQL), Docker, Kubernetes, CI/CD, Azure DevOps, Jenkins, GitHub Actions
Engineering Practices: Microservices Architecture, OIDC/OAuth2, xUnit, Postman, Agile/Scrum, Code Review, Query Optimization

Projects

LedgerFlow | *C#, ASP.NET Core, EF Core, SQL Server, AKS*

- Built a **microservices**-based transaction platform using **ASP.NET Core Web API** and **Entity Framework Core**, deploying containerized services to **Azure Kubernetes Service** with **Azure DevOps** CI/CD pipelines and **xUnit** test coverage.
- Implemented **OIDC**-based authentication and a React/TypeScript dashboard consuming REST APIs backed by **SQL Server**, applying stored procedures and query tuning for real-time transaction reporting across multi-account banking workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a scalable fintech platform on **Kubernetes** with Next.js, TypeScript, and MongoDB, supporting 50,000+ users and 1M+ transactions via Plaid bank-linking and token-based firewall security across real-time filtering workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an AI-powered SQL development workspace on the Chat2DB codebase across Java and TypeScript, supporting 10+ databases including **SQL Server** and PostgreSQL with Docker-based deployment workflows for database management.